

Welcome to the workshop!

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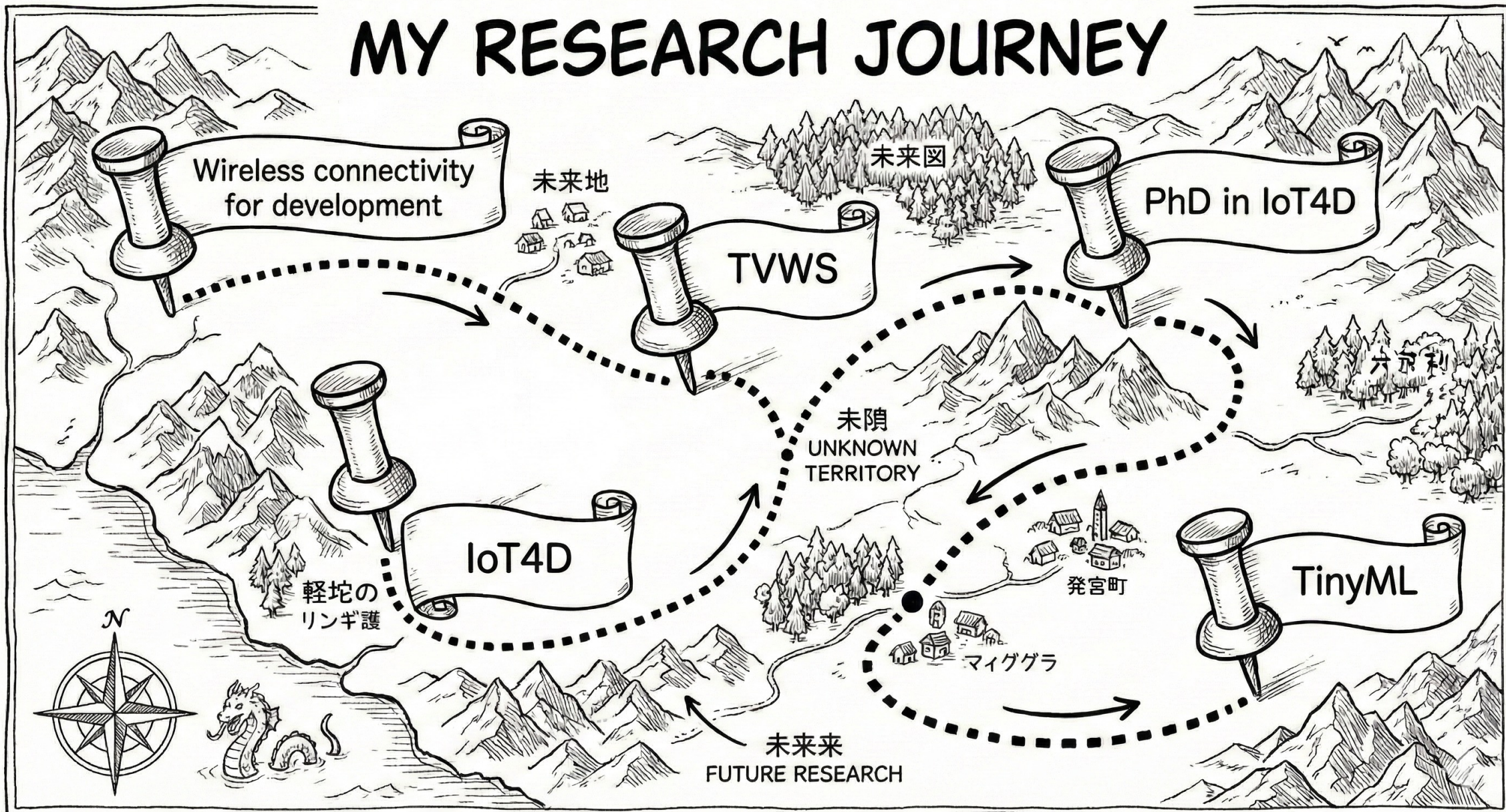
Visiting Professor at Kobe Institute of Computing (KIC) in Kobe, Japan

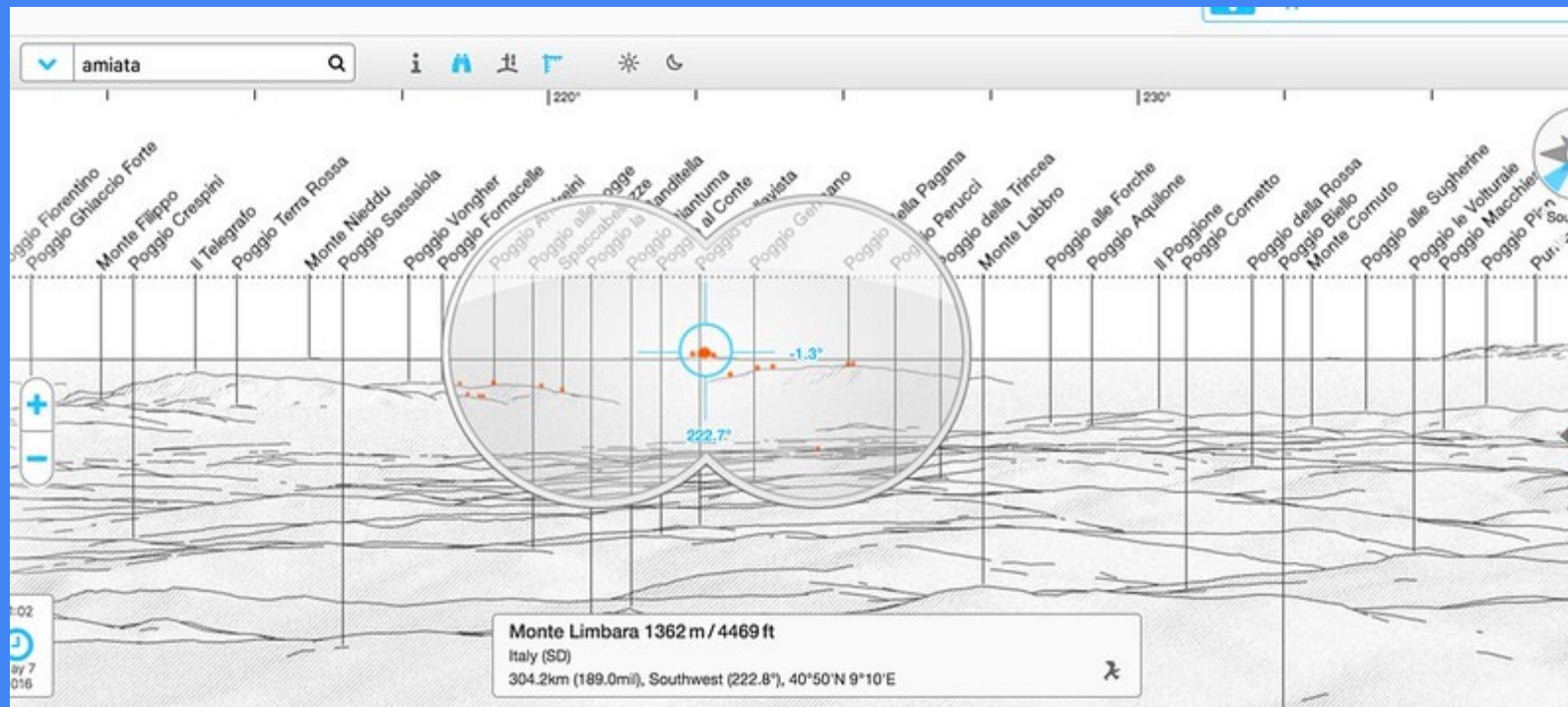
AIEng4D Co-Chair

IEEE Senior Member and an ACM Senior Member



MY RESEARCH JOURNEY



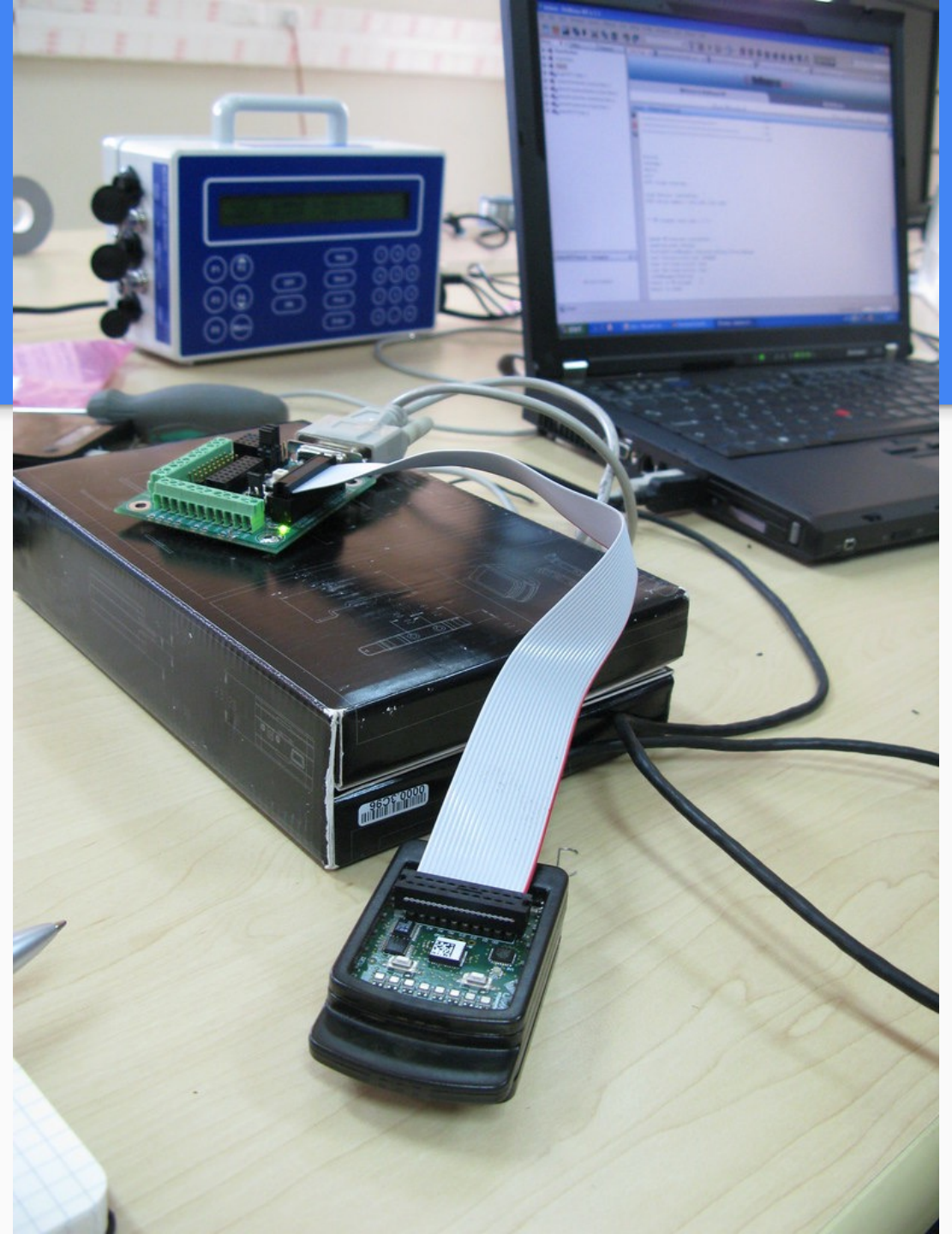




IoT4D



IoT4D

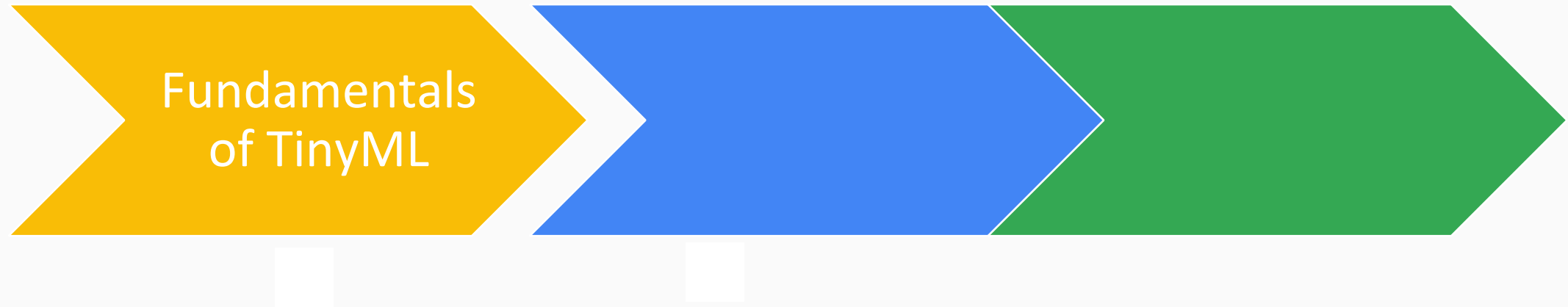




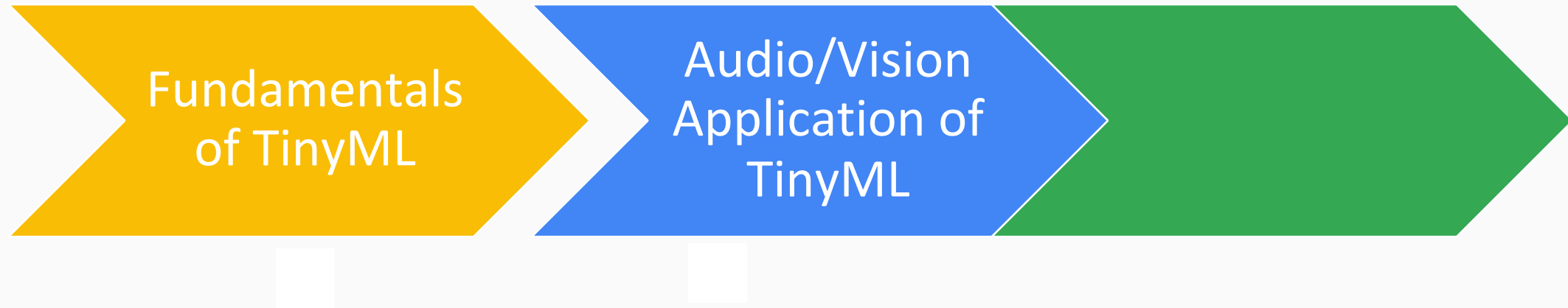


What will we learn?

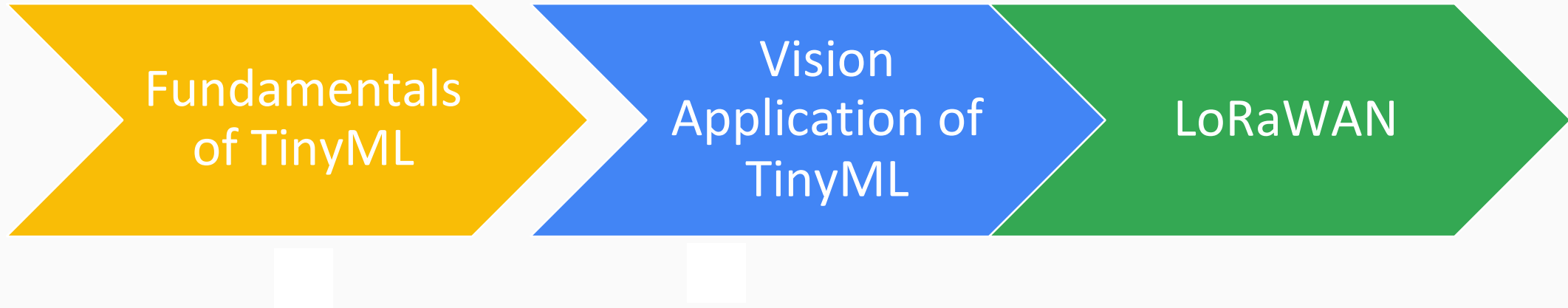
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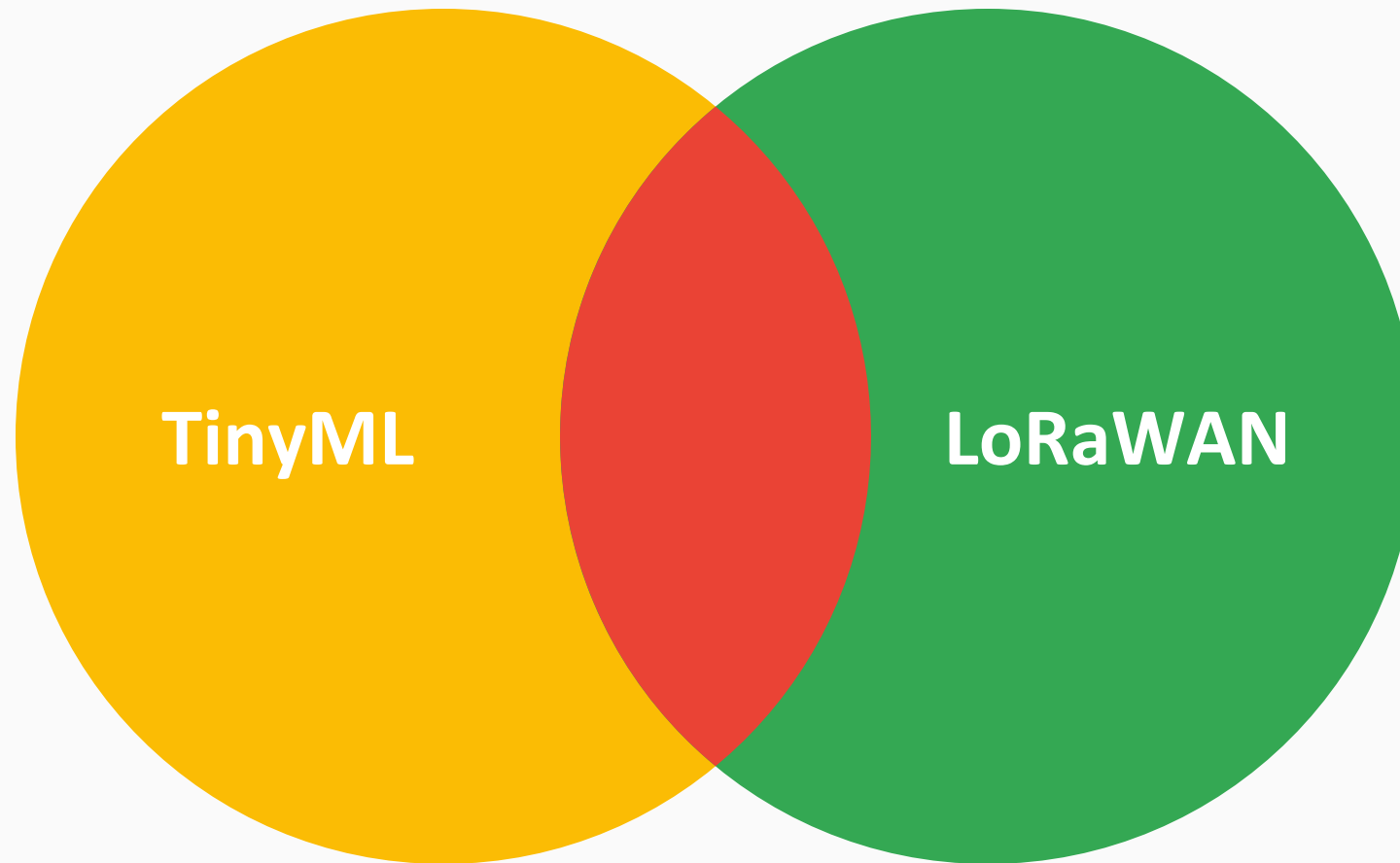
What will We learn?



What will We learn?



Where TinyML meets LoRaWAN



Where TinyML meets LoRaWAN

CAMARILLO, Calif., Sept. 22, 2026 – [Semtech Corporation](#) (Nasdaq: SMTC), a leading provider of high-performance semiconductors powering AI data center networking and intelligent, connected Internet of Things (“IoT”) devices worldwide, today announced its [LoRa Plus™ LR2021](#) transceiver incorporating fourth-generation LoRa® IP was named a winner of the 2026 EDGE Award for Design Innovation. The EDGE Awards are presented by the Engineering Design & Automation Group of Endeavor Business Media to champion innovative technologies launched in the past year.

The 2026 EDGE Awards celebrate outstanding innovation in product design and function for the engineering industry. The LR2021 was named a winner in the Communications & Networks category. This award recognizes an innovation that received the most votes in a category from a Readers’ Choice poll.

The LR2021 was also recently selected for the 25th IoT Gold Award – Innovative Product Award at IoT Shenzhen 2026.

The Technology Behind FLRC

Fast Long-Range Communication (FLRC), featured in the LR2021, offers an enhanced link budget and significantly higher data throughput. By utilizing coherent Minimum-Shift Keying (MSK) modulation, FLRC achieves long-range transmission with notable spectral and energy efficiency. Unlike LoRa, FLRC does not employ spread spectrum methods; consequently, the raw bit rate is equivalent to the channel bandwidth prior to the application of Forward Error Correction (FEC). This direct mapping enables FLRC to deliver higher throughput while maintaining the extended range that long-range applications require.

Key Advantages of FLRC

- **High Data Throughput:** Supporting data rates up to 2.6Mbps, FLRC enables applications requiring significant volumes of data transfer, including images, audio, extensive data logging, Firmware Updates Over-The-Air (FUOTA), and **Edge** AI model training.
- **Extended Range:** FLRC offers 5dB improved sensitivity compared to Bluetooth Low Energy (BLE), effectively doubling operational range at comparable bit rates. While BLE typically reaches 100 meters, FLRC extends this to 1 kilometer (km) in outdoor environments. Sub-GHz operation delivers superior material penetration, ideal for deep indoor deployments.
- **Low Power Consumption:** Optimized for battery-powered platforms necessitating prolonged runtime. Typical security camera deployments achieve 2+ years of battery life with periodic image transmission.

How are we going to get there?

Edge
Device



& Sensors

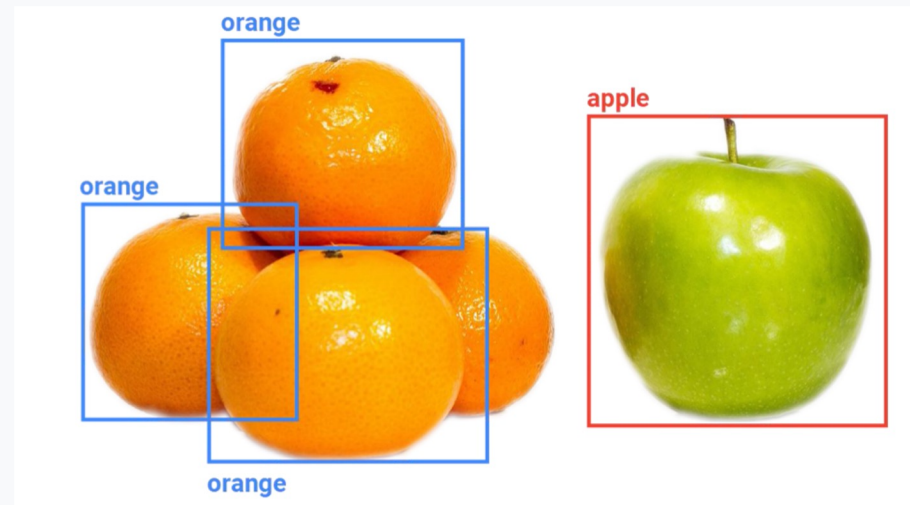


Speech



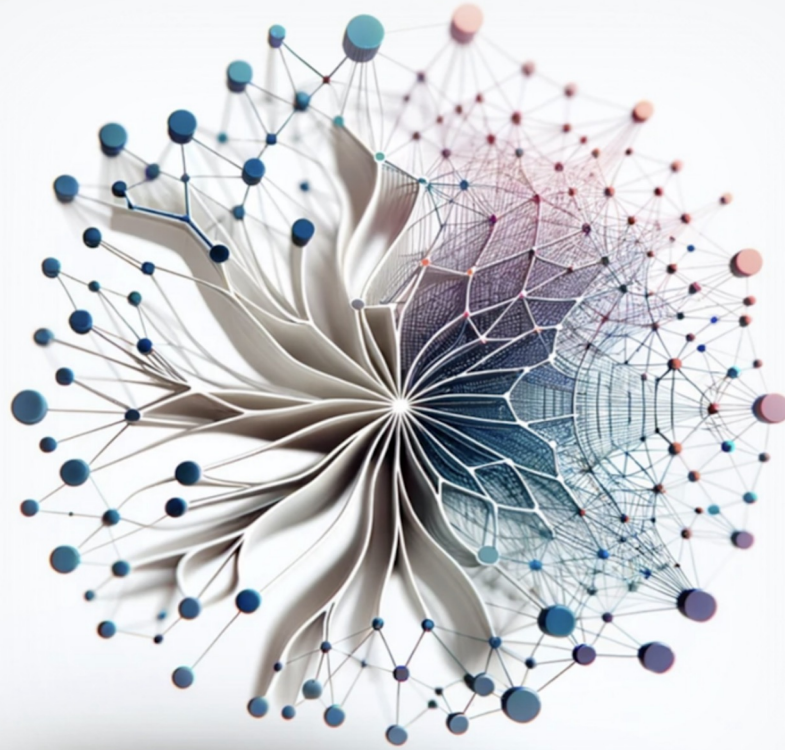
Okay, Google.

Vision



Thank you, Prof. Marcelo Rovai!

Machine Learning Systems with TinyML



Edited by Prof. Vijay Janapa Reddi
Harvard University



https://github.com/harvard-edge/cs249r_book

Tentative Agenda

- **Day 1 — Sense: Machines That See**
 - Intelligence at the Edge: Why TinyML, and Why Now
 - Lab — Nicla Setup Clinic: getting everyone to a blinking LED; First Light: Your Nicla Opens Its Eyes
 - Vision Models on Microcontrollers
- **Day 2 — Transmit: Ten Kilometres on a Coin Cell**
 - The LoRaWAN Story: Long Range, Low Power, Tiny Payloads
 - From Device to Dashboard: Introduction to MQTT
 - Lab — Seeing Your Data: TTN to MQTT to Smartphone

Tentative Agenda

- **Day 3 — Listen and Build**
 - Keyword Spotting and Audio Scene Classification
 - Lab — "Hey Nicla": Building Your Own Wake Word
 - Project Kickoff: Pitch, Team Up, Scope Down
 - Build Session I
- **Day 4 — The New Frontier**
 - When the Edge Learns to Talk: LLMs on the Arduino UNO Q
 - Build Session II
 - Demo Session: Show What You Made
 - Roundtable — What Comes After TinyML? Open Problems and Open Collaboration



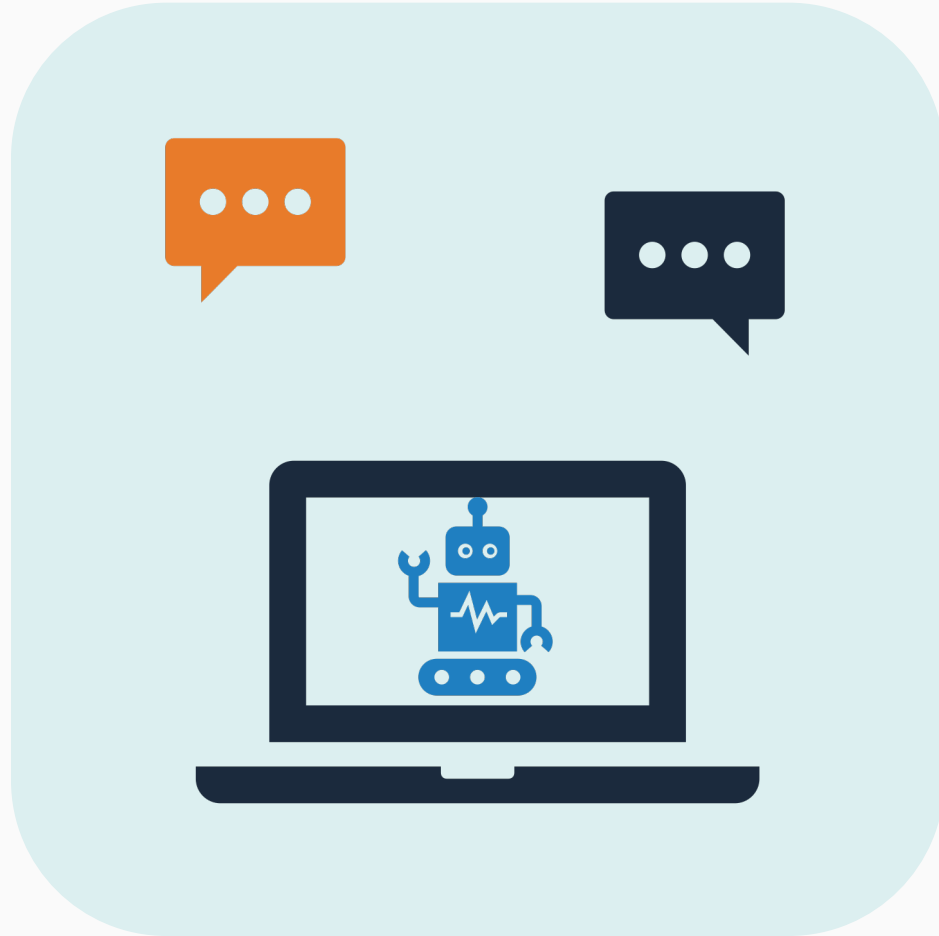
Things will not work (we all have different operating systems, network setups, etc)

We will make things work together.

Please be nice to your colleagues and share your knowledge.

"What we have to learn to do, we learn by doing."

Aristotle



You are more than welcome to use AI tools such as ChatGPT, Claude and Gemini during the labs.

Use them to debug errors, explain code and explore ideas.

Make sure you understand what the code does before you run it.

Training material

HANDS-ON WORKSHOP · IMPERIAL COLLEGE LONDON

Sense, Think, Transmit

TinyML and LoRaWAN for Real-World Deployments

When 12–15 October 2026 **Where** Imperial College London **Instructor** Marco Zennaro, ICTP

[Day 1 · Sense](#) [Day 2 · Transmit](#) [Day 3 · Listen and Build](#) [Day 4 · The New Frontier](#) [All slides](#)

Four days from sensing to sending: running vision and audio models on microcontrollers, getting their results across kilometres with LoRaWAN, and finishing with LLMs on the Arduino UNO Q. Slides are linked next to each session as PDFs.

DAY 1 Monday 12 October

Sense: Machines That See

LECTURE

Intelligence at the Edge: Why TinyML, and Why Now

[Slides](#) PDF · 6.5 MB

LECTURE

Vision Models on Microcontrollers

[Computer vision](#) PDF · 6.0 MB

[Object detection](#) PDF · 1.8 MB

All the training material is openly available here:

<https://marcozennaro.github.io/imperial-2026/>

Please introduce yourself and let us know:

what is your interest in TinyML?

do you have a project/application in mind already?

have you worked with Arduino?